

Eurostat Pesticide Risk Trend Analysis

Data Management Plan - Final Version

Project Title	Eurostat Pesticide Risk Trend Analysis
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Repository	https://github.com/AbdulQuaiyum918/eu-pesticide-risk-trends
Zenodo DOI	https://doi.org/10.5281/zenodo.20467709
DMP DOI	https://doi.org/10.70124/5rmcx-fah87
Template	FWF Stand-alone DMP Template (2022 edition)

Prepared in accordance with the FWF Data Management Plan requirements and the FAIR data principles (Findable, Accessible, Interoperable, Reusable) as defined by GO FAIR.

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1 Data Description and Collection

1.1 What data will be collected or produced?

The project analyses pesticide sales and application risk indicators published by Eurostat under the Farm-to-Fork Strategy monitoring framework. The primary dataset is the Agri-environmental indicator - risk from pesticides (Eurostat code: AEI_FM_SALPEST09), covering EU-27 Member States for reference years 2011-2022. Data are provided in CSV and SDMX formats, with an estimated volume of approximately 50 MB raw and 5 MB processed.

Data types collected:

- Quantitative time-series: annual pesticide sales quantities per country (kg active substance)
- Derived risk-index scores: harmonised risk indicators HRI-1 and HRI-2
- Geospatial country-level polygons (NUTS-0) for choropleth mapping
- Processed analysis outputs: cleaned CSVs, statistical summaries, figures (PNG/SVG)

1.2 Data origin and existing datasets

All source data are retrieved programmatically from the Eurostat REST API and the Eurostat bulk download facility. No personal or sensitive data are collected. The Eurostat datasets are published under the CC BY 4.0 licence, permitting reuse with attribution. No new primary data collection requiring ethical approval is involved.

1.3 Data formats and standards

Source data are retrieved in SDMX-JSON and CSV (UTF-8, comma-separated). Processed outputs are stored as tidy-format CSV files following the Frictionless Data Tabular Data Package specification. Visualisations are exported in SVG (vector) and PNG (300 dpi raster). Metadata are recorded in JSON-LD following the schema.org Dataset vocabulary and DataCite Metadata Schema 4.5.

2 Documentation and Metadata

2.1 Metadata standards

The project applies the DataCite Metadata Schema 4.5 for all archived research outputs on Zenodo. A `codemeta.json` file documents the software environment following the CodeMeta vocabulary. A `README.md` at the root of the repository provides a human-readable data dictionary, provenance narrative, and reproduction instructions.

2.2 Data documentation artefacts

- `data/raw/` - immutable copies of downloaded Eurostat files with SHA-256 checksums
- `data/processed/` - cleaned datasets with a codebook (`codebook.md`) describing every variable
- `notebooks/` - Jupyter notebooks with inline narrative (literate programming approach)
- `CHANGELOG.md` - version history following Keep a Changelog conventions
- `environment.yml` - Conda environment specification for full software reproducibility

3 Data Storage and Security

3.1 Storage during the project

Working data and code are stored in the GitHub repository at <https://github.com/AbdulQuaiyum918/eu-pesticide-risk-trends>, with access restricted to group members during active development. GitHub provides continuous off-site replication. Each member maintains a local clone, providing additional redundancy across at least four independent machines.

3.2 Backup strategy

Commits are pushed to the remote repository after every working session (minimum daily during active development). Large binary assets such as figures and raw CSVs are tracked with Git LFS. A full repository archive is exported to a personal OneDrive folder at each project milestone.

3.3 Security and access control

No personal or confidential data are handled; all source data are openly licensed. GitHub repository access uses SSH key authentication. Secrets and API tokens are stored in environment variables and excluded from version control via `.gitignore`.

4 Legal and Ethical Requirements

4.1 Intellectual property and licensing

Source datasets are published by Eurostat under the Creative Commons Attribution 4.0 International (CC BY 4.0) licence. All derived project outputs - code, processed data, figures, and documentation - are

released under the MIT Licence (code) and CC BY 4.0 (data and reports), as declared in the repository LICENSE file and the Zenodo metadata record.

4.2 Ethical considerations

The project does not involve human participants, personal data, sensitive ecological data, or proprietary datasets. No ethics committee approval is required. All data sources are cited in accordance with Eurostat's reuse policy.

4.3 GDPR compliance

No personal data within the meaning of the EU General Data Protection Regulation (2016/679) are processed. This section is therefore not applicable to the present project.

5 Data Sharing and Open Access

5.1 Data sharing and open access

All project outputs are published openly in accordance with the FWF Open Access Policy and the FAIR principles. The final dataset, analysis notebooks, and documentation are archived on Zenodo (a CERN-operated, OpenAIRE-compliant repository) with a persistent DOI: <https://doi.org/10.5281/zenodo.20467709>. The GitHub repository remains public for ongoing community access and issue tracking.

5.2 Restrictions on sharing

No restrictions on sharing apply. All outputs are made available immediately upon project completion with no embargo period.

5.3 Making data findable

Findability is ensured through: (a) a persistent DOI registered with DataCite; (b) rich metadata on Zenodo indexed by OpenAIRE and Google Dataset Search; (c) a CITATION.cff file in the repository enabling automated citation generation; (d) keywords aligned with the AGROVOC controlled vocabulary for agricultural data.

6 Long-Term Preservation

6.1 Long-term archiving

Zenodo provides long-term preservation (minimum 20 years) backed by CERN infrastructure, including geographic replication and format migration policies. The repository holds CoreTrustSeal certification as a trustworthy digital repository. All archived files include MD5 checksums to support future integrity verification.

6.2 Formats chosen for preservation

Data are archived in CSV (UTF-8) - a non-proprietary, widely supported, plain-text format recommended by the Library of Congress and the UK Digital Curation Centre for long-term tabular data preservation. Figures are archived in SVG (lossless vector) alongside PNG rasters. Notebooks are archived in .ipynb format and as static HTML exports.

6.3 Responsible person for data management

Md Abdul Quaiyum (12449471) serves as the designated data steward and is responsible for maintaining the Zenodo record, responding to data access queries, and applying any required corrections via Zenodo versioning.

7 Resources and Responsibilities

7.1 Resources for data management

All tools and platforms used are free and open-source or freely available to researchers: Python 3.11 (Anaconda distribution), Jupyter Lab, pandas, matplotlib, Git/GitHub, and Zenodo. No additional budget is required for data management activities. Storage costs are covered by free-tier allocations (GitHub: unlimited public repositories; Zenodo: 50 GB per record).

7.2 DMP maintenance

This DMP was drafted using the FWF Stand-alone DMP template and will be reviewed at each project milestone and updated as necessary. The machine-actionable version is maintained on the DMP platform with DOI: <https://doi.org/10.70124/5rmcx-fah87>.